

SKIN SCAN

AI-Powered Skin Disease Detection

OUR TEAM

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RESEARCH

- Conducted comprehensive research on existing AI-based dermatology tools.
- Explored state-of-the-art techniques, including Convolutional Neural Networks (CNNs) for skin disease classification.
- Investigated applications of AI in healthcare, such as precision medicine and dermatological image analysis.
- Compiled insights into cutting-edge models like DenseNet and Inception architectures for medical image processing.



RESEARCH FINDINGS

- Many AI tools achieve high accuracy but lack generalizability.
- Performance often varies across diverse skin tones and conditions due to biased datasets.
- Most tools focus on specific diseases (e.g., melanoma) rather than comprehensive classification.



DATA COLLECTION AND ANALYSIS

- Dataset Source : [kaggle](#)

- DATASET OVERVIEW :

Number of Classes:

9 distinct skin disease categories.

Classes Included:

Actinic Keratosis

Atopic Dermatitis

Benign Keratosis

Dermatofibroma

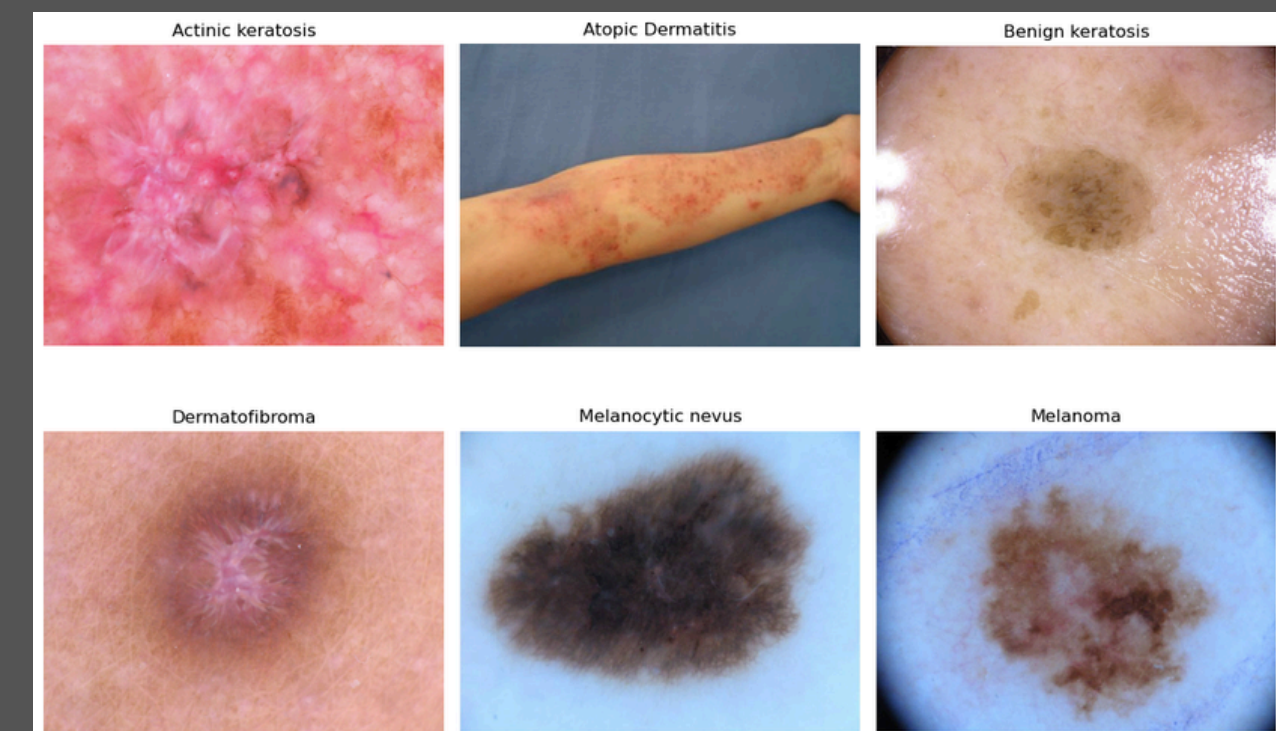
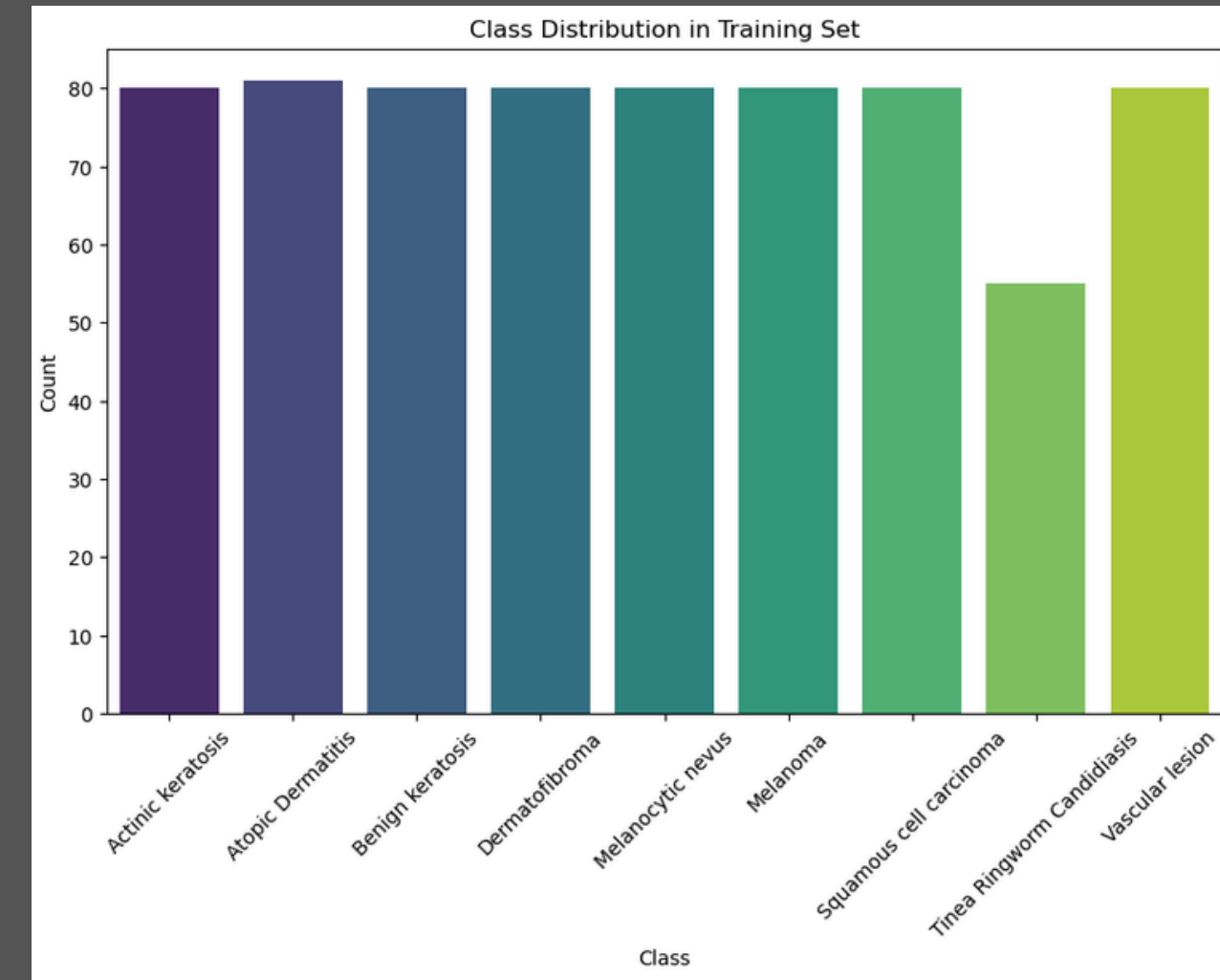
Melanocytic Nevus

Melanoma

Squamous Cell Carcinoma

Tinea (Ringworm, Candidiasis)

Vascular Lesion



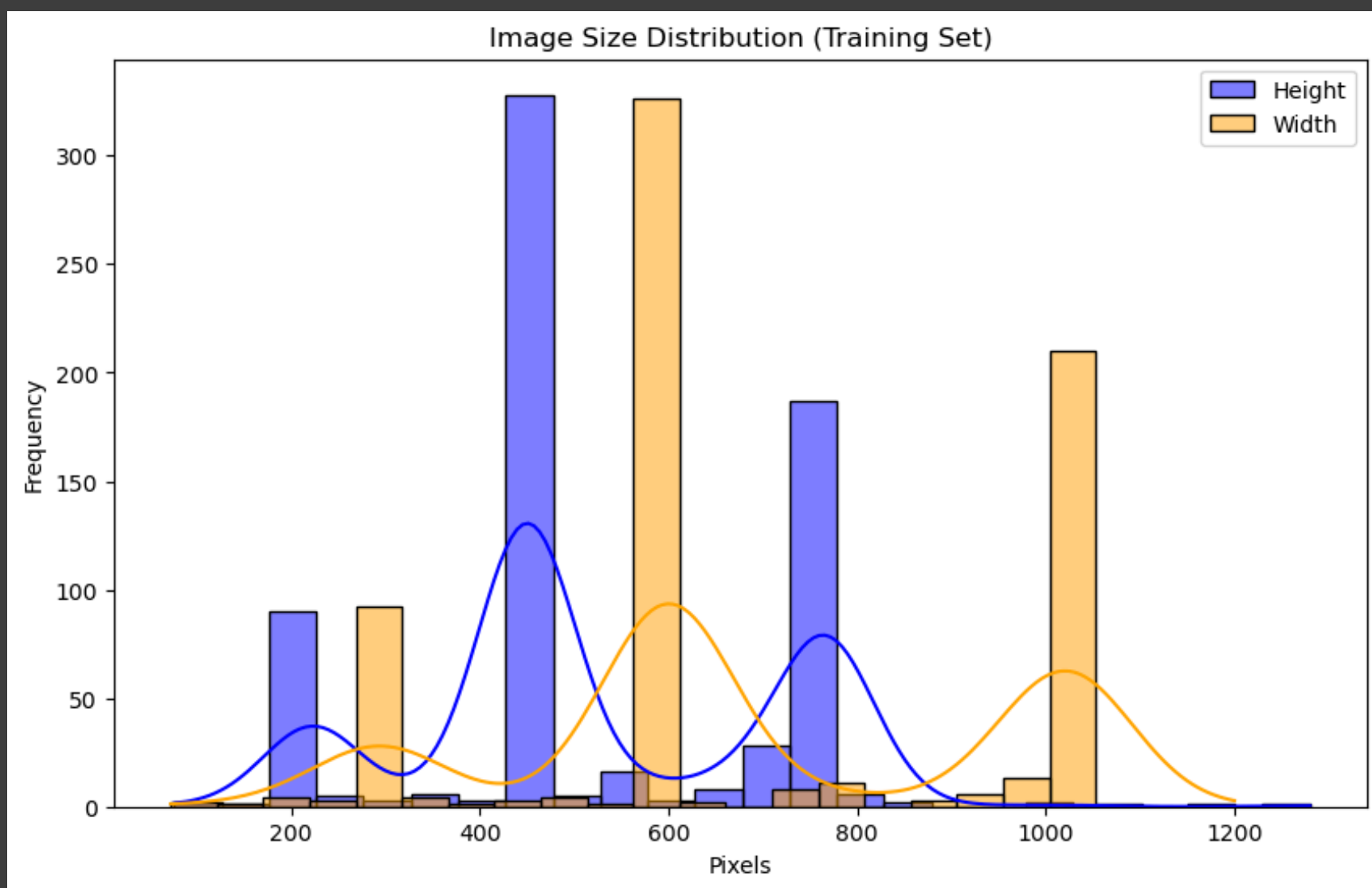
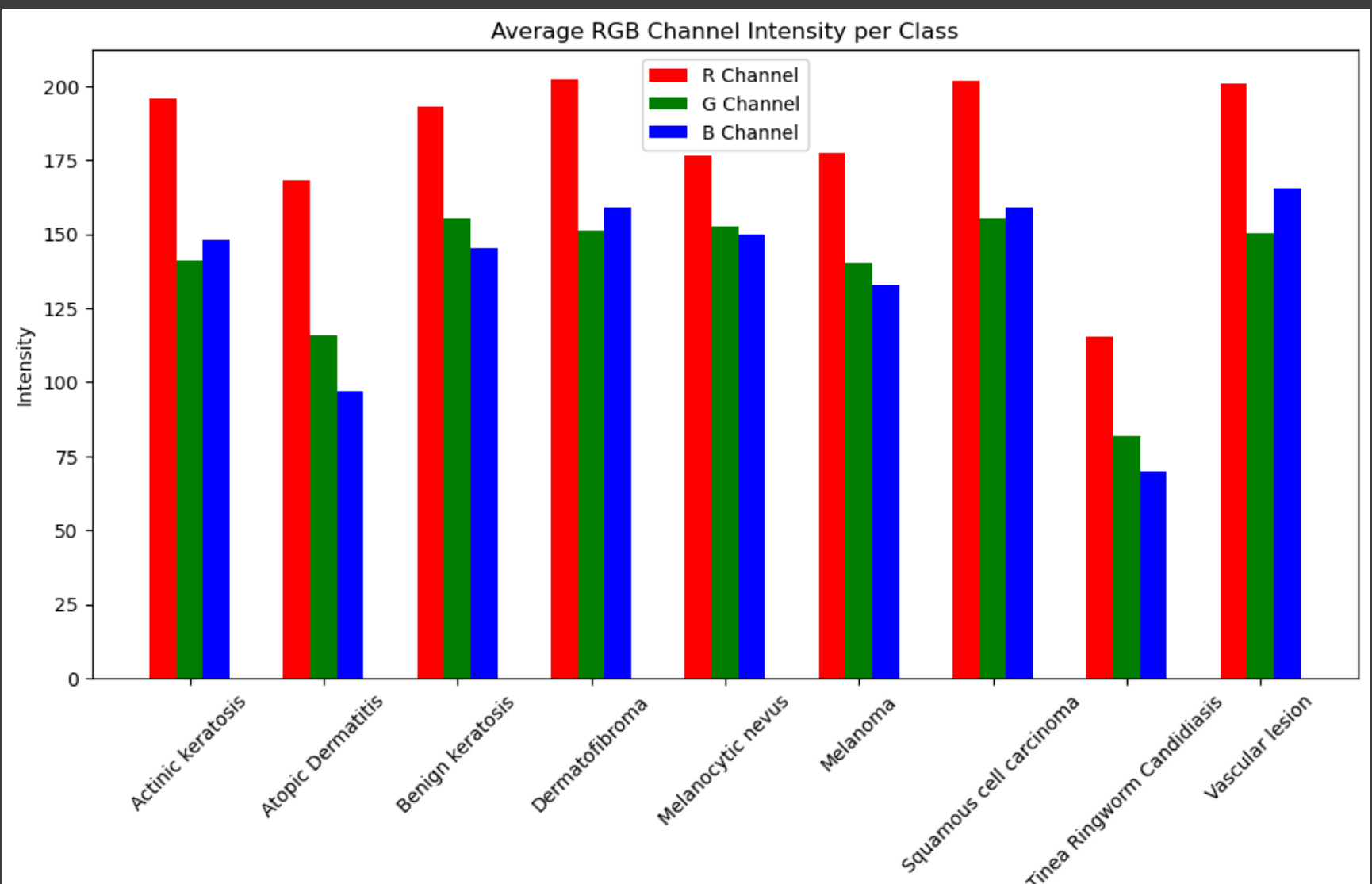


IMAGE SIZE DISTRIBUTION

Dominant Image Heights:
around 400 pixels and 600 pixels

Dominant Image Widths : :

around 400 pixels and 1000
pixels



RGB CHANNEL INTENSITY PER CLASS

RED CHANNEL DOMINANCE

VARIATION ACROSS CLASSES

POTENTIAL FOR DIFFERENTIATION

DATA AUGMENTATION

Techniques Used :

Rotation (rotation_range=20): Randomly rotates images up to 20 degrees.

Width/Height Shifting (width_shift_range=0.2, height_shift_range=0.2): Moves images horizontally or vertically by up to 20%.

Shear Transformation (shear_range=0.2): Applies random shearing effects.

Zooming (zoom_range=0.2): Randomly zooms into images by up to 20%.

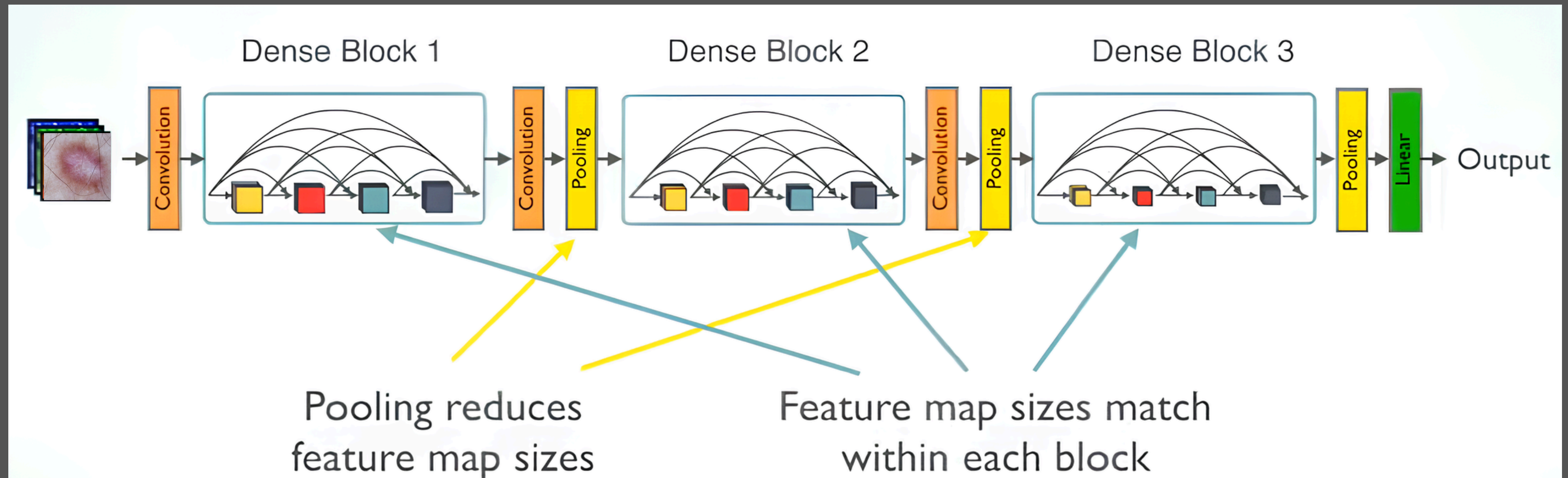
Horizontal Flipping (horizontal_flip=True): Flips images horizontally.



MODEL BUILDING

Model used : DenseNet-121

- used DenseNet-121, a pre-trained convolutional neural network (CNN) architecture, as the backbone for feature extraction.

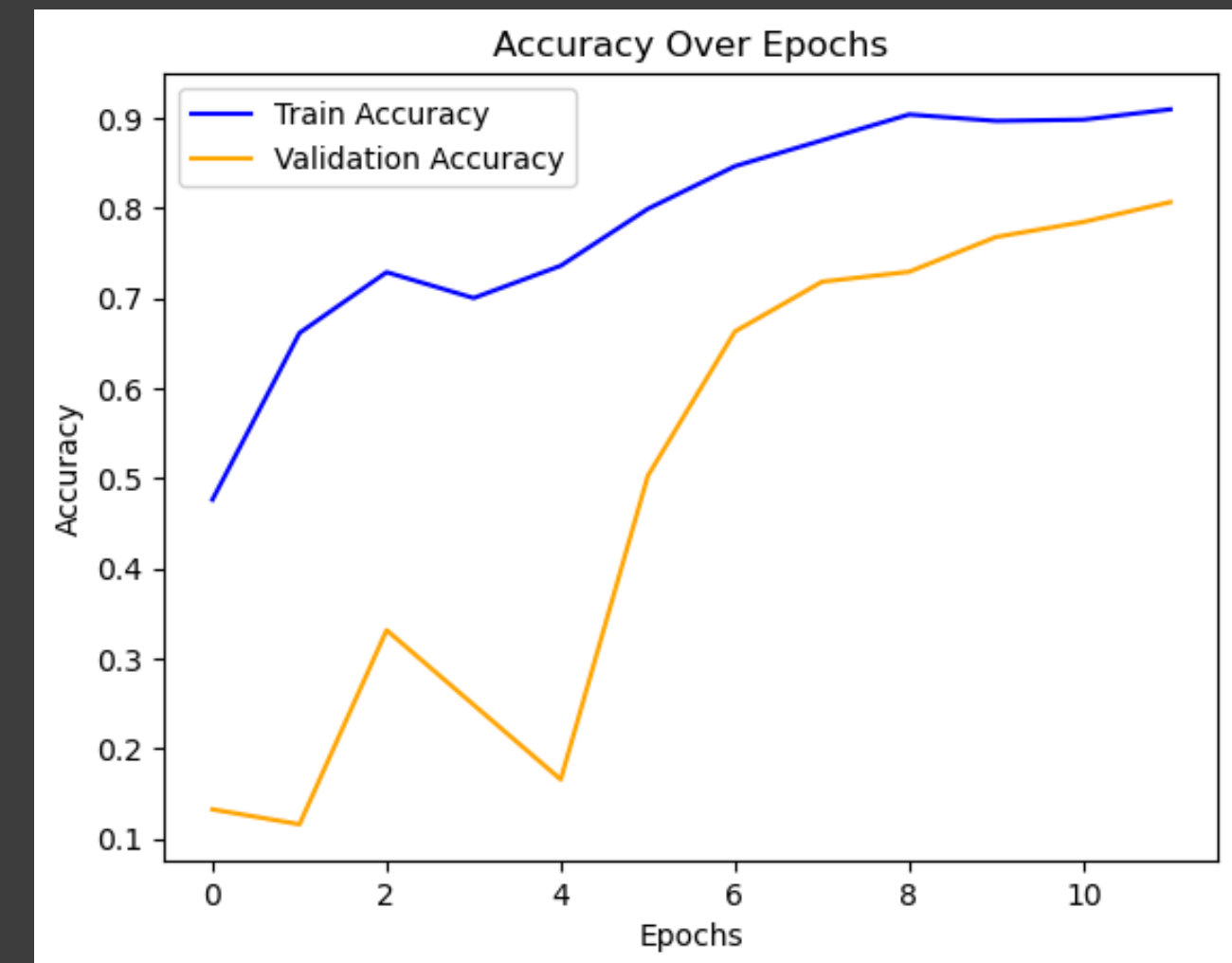
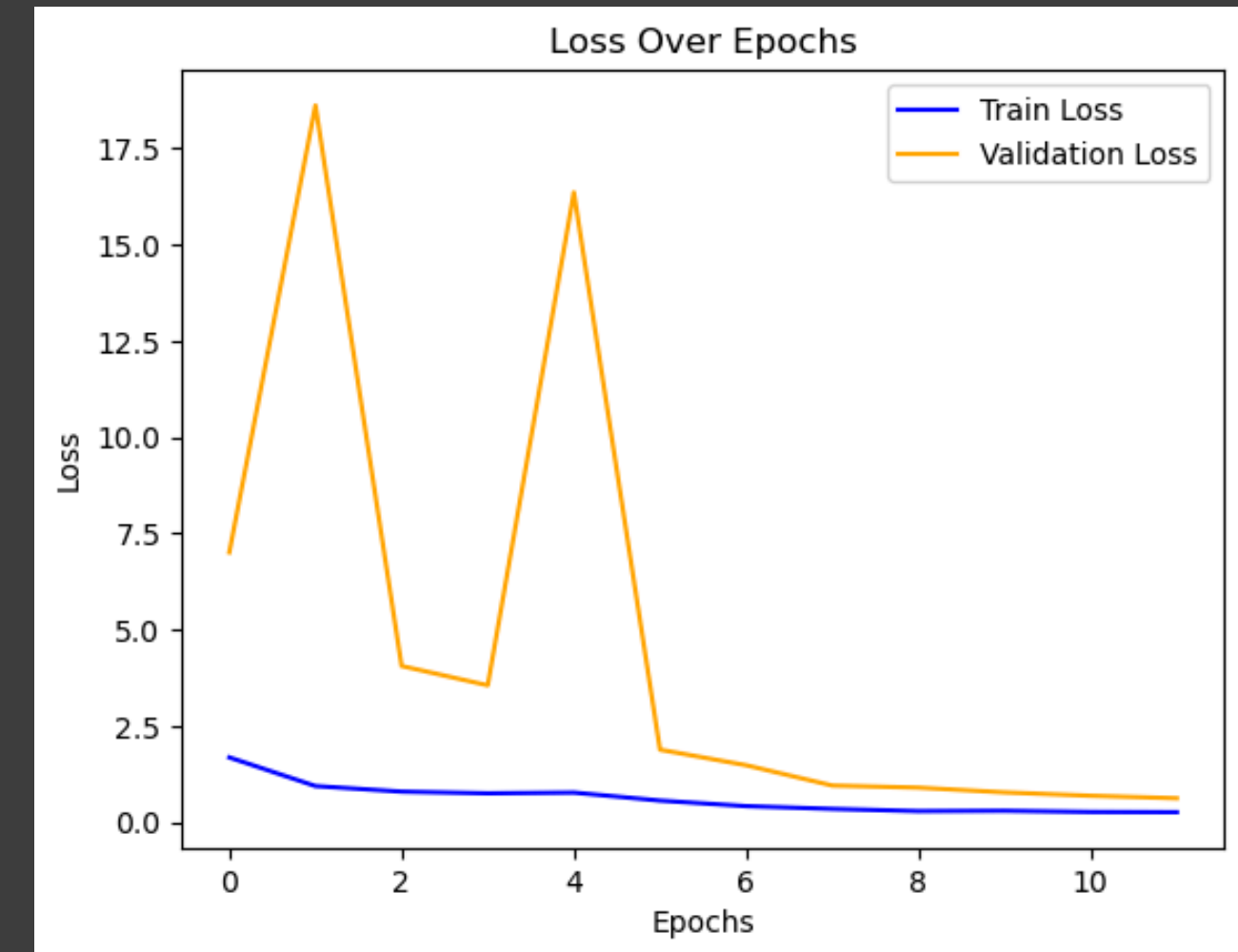


RESULTS OBTAINED

Training accuracy obtained : **0.7971 (79.71%)**

Validation accuracy obtained : **0.8066 (80.66%)**

1. **Accuracy Graph:** The training and validation accuracy steadily improve over epochs, with validation accuracy reaching 80.66%, indicating effective learning and good generalization to unseen data.
2. **Loss Graph:** The training loss decreases consistently, while validation loss shows an initial sharp drop and stabilizes, demonstrating the model's ability to minimize error without overfitting.



MODEL DEPLOYMENT



Key Features:

- **Flask Framework:** Used to create a web application for skin disease classification.
- **Machine Learning Model Integration:** TensorFlow model (skin_disease_model.h5) predicts skin conditions based on uploaded images.
- **Routes:**
 - `/`: Main route for uploading and analyzing images.
 - `/history`: Displays analysis history.
 - `/api/export/<analysis_id>/pdf`: Exports analysis results as a PDF.
- **Database:** SQLite database (skin_analysis.db) stores analysis history and disease information.
- **Image Processing:** Preprocesses uploaded images using `load_img` and `img_to_array`.

MODEL DEPLOYMENT

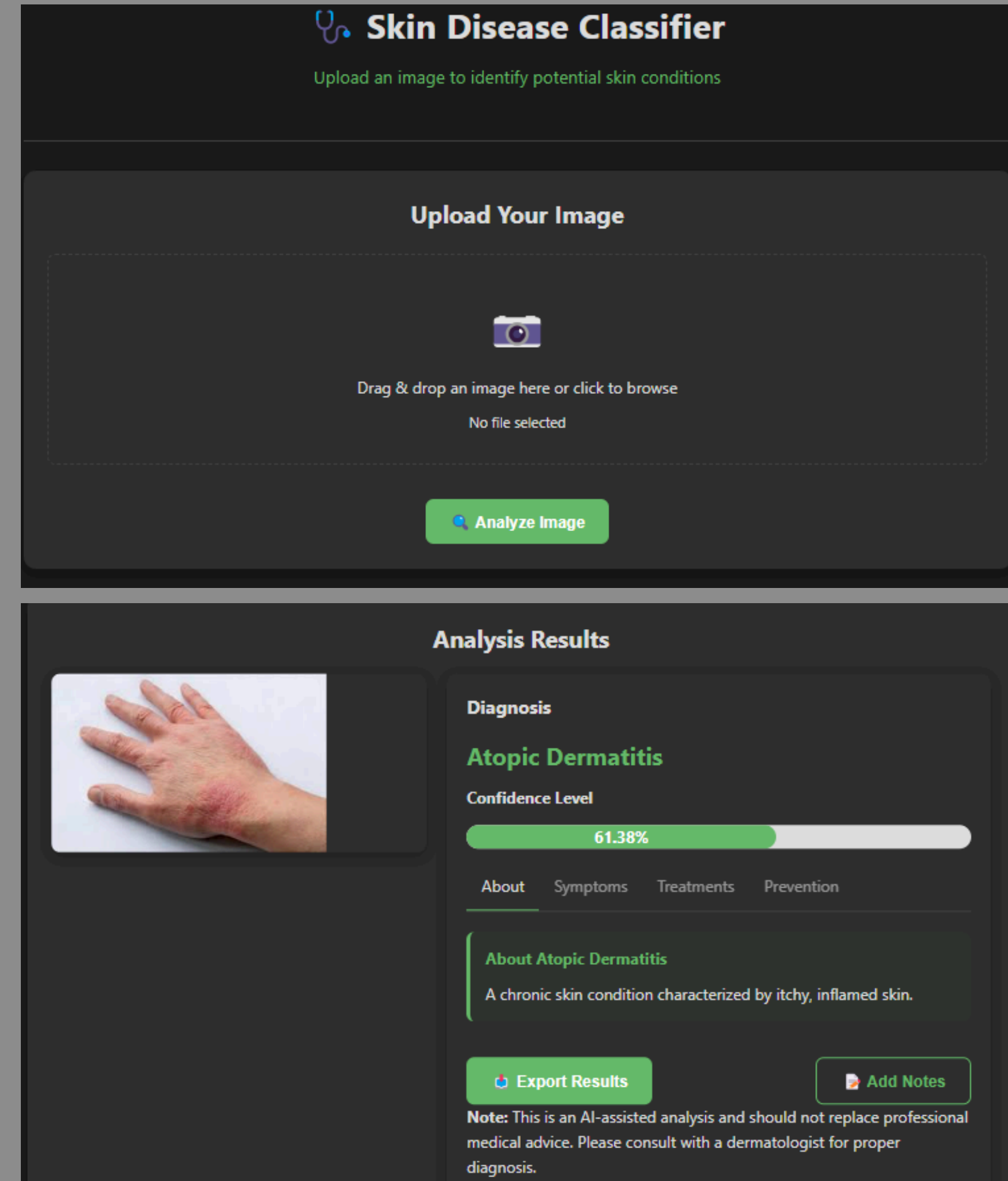
Frontend Design for Skin Disease Classifier

HTML Features:

- **File Upload Form:** Users can upload images for analysis.
- **Results Display:** Shows disease prediction, confidence level, and detailed information.

CSS Highlights:

- **Responsive Design:** Optimized for desktop view.
- **Theme Toggle:** Light/Dark mode functionality using CSS variables.
- **Modern UI Elements:** Styled buttons, cards, and file upload area with hover effects.



PROJECT OVERVIEW & STRATEGIC INSIGHTS

Objective:

To develop a machine learning model capable of accurately classifying multiple skin diseases using dermatoscopic images.

Team Contributions:

- Researched state-of-the-art AI and dermatology solutions.
- Focused on practical implementation using DenseNet-121 CNN architecture.
- Addressed dataset imbalance and image variability via augmentation techniques

Key Challenges Identified:

- Dataset bias impacting generalizability across skin tones.
- Existing models are disease-specific; lack comprehensive classification capabilities.

Implementation Summary & Performance

Dataset Summary:

- Source: Public dermatology image repository
- Classes: 9 (including Melanoma, Dermatitis, Tinea, etc.)
- Preprocessing: Normalization, image resizing, and augmentation

Model Strategy:

- Architecture: DenseNet-121 (pre-trained CNN)
- Augmentation Techniques: Rotation, zoom, shear, flipping, shifting
- Training accuracy obtained : 0.7971 (79.71%)
- Validation accuracy obtained : 0.8066 (80.66%)

Next Steps:

- Enhance dataset diversity
- Improve model robustness across demographics
- Explore web/mobile deployment for end-user accessibility

CONCLUSION

We successfully built a skin disease classification model using DenseNet-121 and enhanced it with data augmentation and transfer learning techniques to improve its performance. The model was trained and validated effectively, achieving a decent accuracy, and was deployed using Flask, making it accessible for real-time predictions through a web application.